

Kalgoorlie AMC Refinery Expansion

Economic Contribution of AMC

Without Expansion

	1	2	3	4	5	6	7	8	9	10	11
(y) Year	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
(c) Contribution (millions of \$)	250	500	750	1000	1250	1500	1750	2000	2250	2500	2750

Contribution = \$250 million per year

With Expansion

Contribution = 250y

				1	2	3	4	5	6	7	8
(y) Year	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
(c) Contribution (millions of \$)	250	500	750	1250	1750	2250	2750	3250	3750	4250	4750

Contribution = 500x year + 750

Gold Production of AMC

Without Expansion

starting from the year 2023 as 1
C = 500y + 750 from 2023 as Year 1

	1	2	3	4	5	6	7	8	9	10	11	12
(y) Year	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
(P) Production ('000 tonnes)	20	40	60	80	120	120	140	160	180	200	220	240

Production = 20x year with 2019 as Year 1.

P = 20y

With Expansion

					1	2	3	4	5	6	7	8
(y) Year	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
(P) Production ('000 tonnes)	20	40	60	80	120	160	200	240	280	320	360	400

Production = 40x year + 80 with 2023 as Year 1

P = 40y + 80

Employment Changes

Current (2020)

2800 Western Australians employed.

600 direct: Cost of Wages = \$35000 x 600
= \$21 000 000

2200 indirect: Cost of Wages = \$28000 x 2200
= \$61 600 000

Total wages = \$21 000 000 +
\$61 600 000
= \$82 600 000

During Construction (2021-2022)

2800 + 300 + 400 = 3500 Western Australians employed.

Cost of Wages of Construction Workers = \$30000 x 400
= \$12 000 000

Cost of Extra indirect employees = \$28000 x 300
= \$8 400 000

Total wages = \$82 600 000
+ \$12 000 000
+ 8 400 000
= \$103 000 000

Expansion Complete & Operating

$2800 + 200 + 150 = 3150$ Western Australians employed

Direct workers: $800 \times \$35000$

$= \$28\,000\,000$

Indirect Workers $= 2350 \times \$28000$

$= \$65\,800\,000$

Total Wages: $\$28\,000\,000$

$+ \$65\,800\,000$

$\$93\,800\,000$

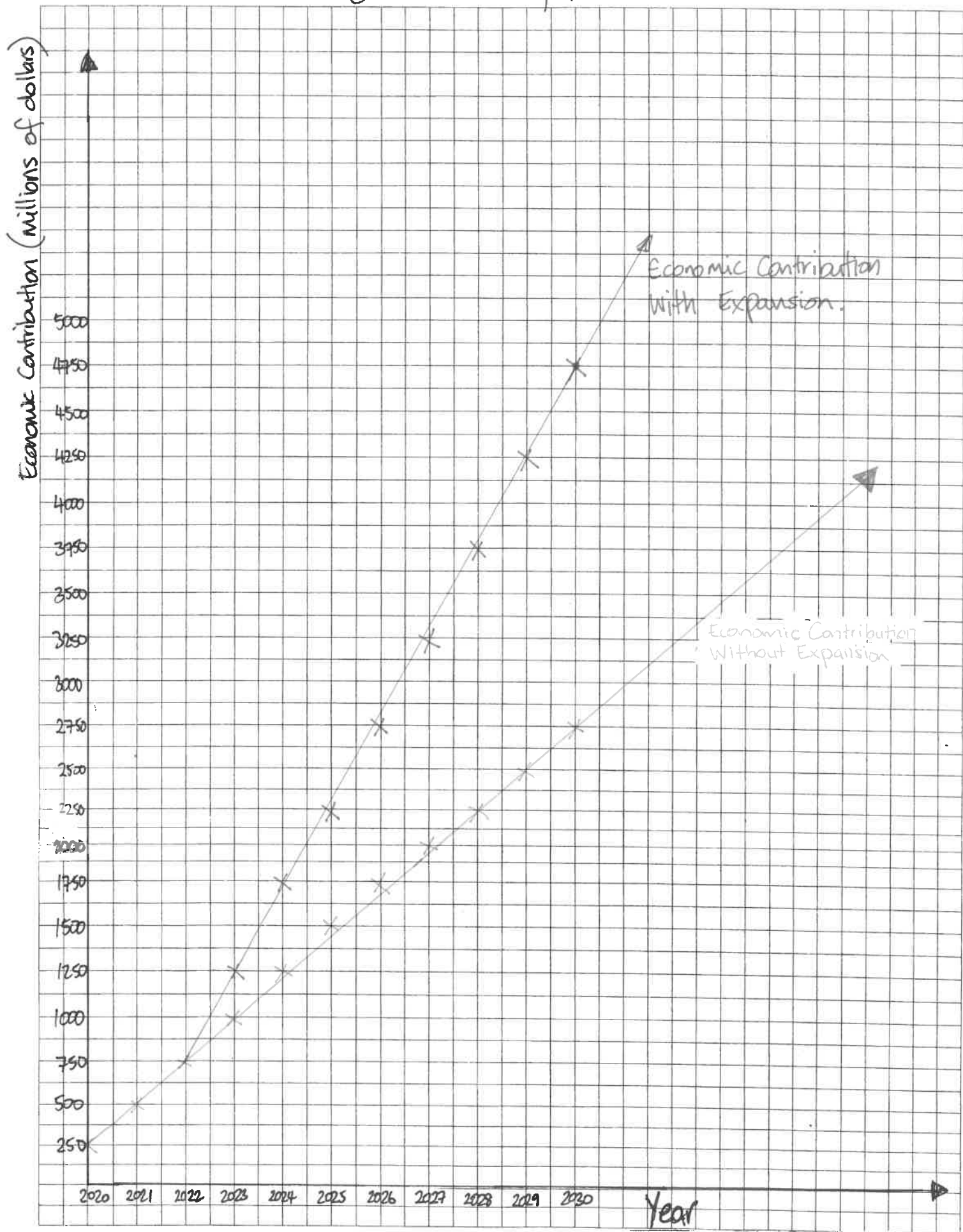
Employment Changes Due to Expansion.

During the construction phase, the number of employees increase from 2800 to 3500 resulting in a wage increase from \$82.6 million to \$103 million. Once construction has completed, an increased workforce is needed for the refinery's operations of 3150. The wage cost decreases from the construction phase to \$93.8 million. Overall there is an increase of the number of people employed in Kalgoorlie.

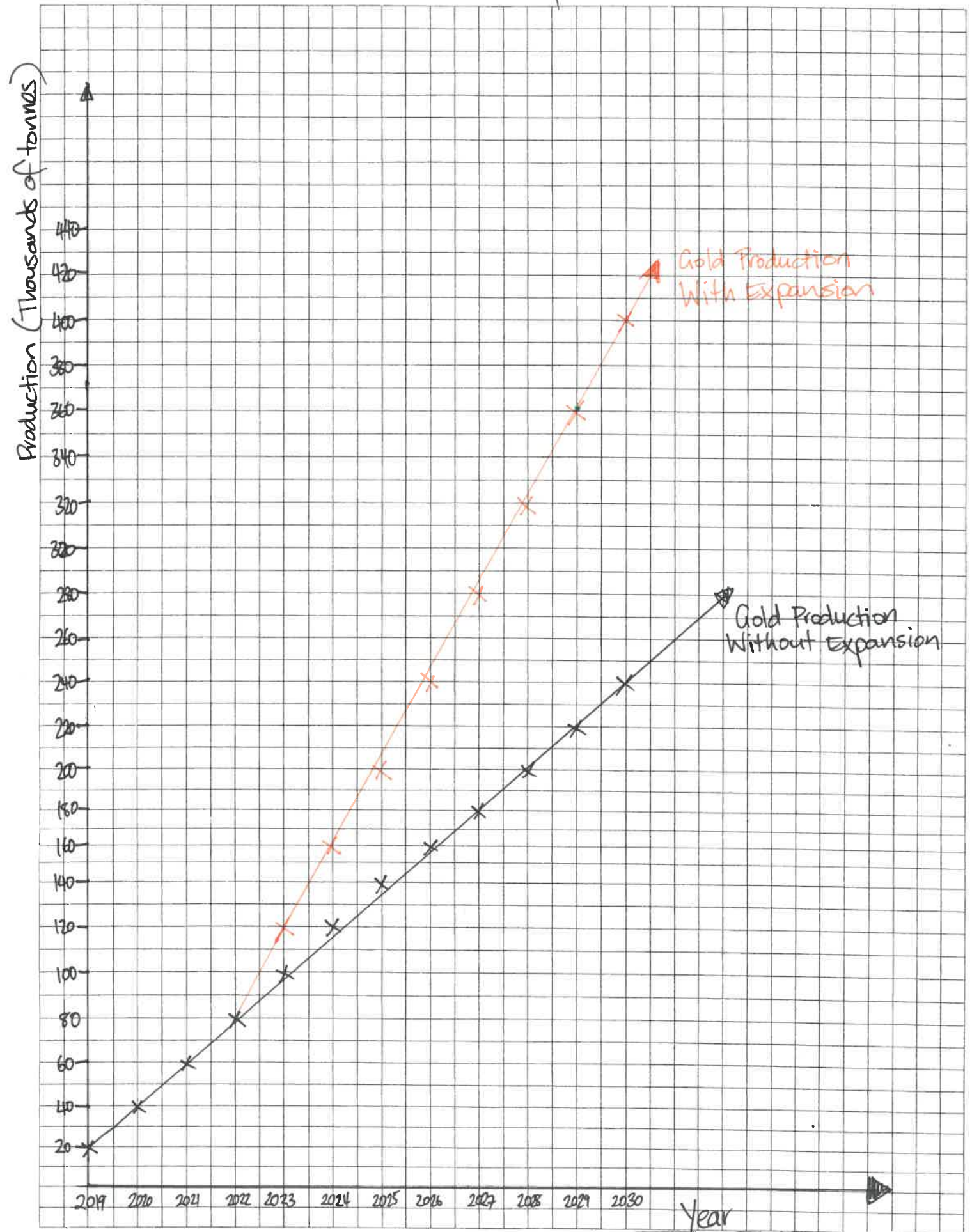
Discussion & Conclusion

There is an increase of employment in Kalgoorlie from 2800 to 3150 in the long term. This will be the case as long as production of gold remains at 40000 tonnes with an increasing gold price. The line of best fit predicts the gold price will continue to increase but the increase from year to year is not always the same. In some years the gold price decreases. At a production of 40000 tonnes, the gold price needs to be at a minimum of \$2345 per tonne / \$2.35 per kg / \$0.0023 per g, which is much less than any of the gold price over the past 10 years. The expansion of AMC's refinery should go ahead as the costs of running the operation and paying wages will be able to be paid. It is unlikely ~~for~~ that the gold price will fall to a low price of \$2345 per tonne.

Economic Contribution of AMC



Gold Production of AMC



Gold Prices from 2010-2020.

